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RECOMMENDED FORMULAS IN SPREADSHEETS USING LEARNED TABLE REPRESENTATIONS

Abstract

The present disclosure relates to methods and systems that automatically identify similar spreadsheets to target spreadsheets. The methods and systems automatically predict formulas that users want to author in a target cell of a target spreadsheet by identifying a reference formula from a similar region in a similar reference sheet that is similar to the target cell. The methods and systems generate a predicted formula by adapting parameters of a reference formula to a context of the target cell. The methods and systems provide an output with the predicted formula in the target cell of the target spreadsheet.

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Background/Summary

BACKGROUND

[0001] Spreadsheets are a popular end-user programming tools for tables, which blend the power of formula-based computation, with an intuitive table-based interface. Today, spreadsheets are used by billions of non-technical users to manipulate tables, most of whom are neither database experts nor professional programmers. Authoring complex formulas remains challenging, as non-technical users need to look up and understand non-trivial formula syntax.

BRIEF SUMMARY

[0002] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0003] Some implementations relate to a method. The method includes automatically generating, by a machine learning model, training data with positive examples of similar spreadsheets and negative examples of dis-similar spreadsheets. The method includes using the training data, by a first machine learning model, to determine whether a pair of spreadsheets are similar. The method includes using the training data, by a second machine learning model, to identify similar spreadsheet regions between the pair of spreadsheets. The method includes providing a first output indicating the pair of spreadsheets are similar or dis-similar. The method includes providing a second output indicating whether the spreadsheets regions are similar or dis-similar between the pair of spreadsheets.

[0004] Some implementations relate to a device. The device includes a processor; memory in electronic communication with the processor; and instructions stored in the memory, the instructions being executable by the processor to: automatically generate, by a machine learning model, training data with positive examples of similar spreadsheets and negative examples of dis-similar spreadsheets; use the training data, by a first machine learning model, to determine whether a pair of spreadsheets are similar; use the training data, by a second machine learning model, to identify similar spreadsheet regions between the pair of spreadsheets; provide a first output indicating the pair of spreadsheets are similar or dis-similar; and provide a second output indicating whether the spreadsheets regions are similar or dis-similar between the pair of spreadsheets.

[0005] Some implementations relate to a method. The method includes receiving a target sheet with a target cell. The method includes identifying, using a first machine learning model, a reference sheet similar to the target sheet. The method includes identifying, using a second machine learning model, a similar region in the reference sheet that is similar to the target cell. The method includes identifying, using the second machine learning model, a reference formula in the similar region in response to receiving an indication a formula is being created in the target cell. The method includes modifying, using the second machine learning model, parameters of the reference formula to a context of the target sheet. The method includes generating, using the second machine learning model, a predicted formula with the modified parameters of the reference formula. The method includes providing an output in the target cell with the predicted formula.

[0006] Some implementations relate to a device. The device includes a processor; memory in electronic communication with the processor; and instructions stored in the memory, the instructions being executable by the processor to: receive a target sheet with a target cell; identify, using a first machine learning model, a reference sheet similar to the target sheet; identify, using a second machine learning model, a similar region in the reference sheet that is similar to the target cell; identify, using the second machine learning model, a reference formula in the similar region in response to receiving an indication a formula is being created in the target cell; modify, using the

second machine learning model, parameters of the reference formula to a context of the target sheet; generate, using the second machine learning model, a predicted formula with the modified parameters of the reference formula; and provide an output in the target cell with the predicted formula.

[0007] Additional features and advantages will be set forth in the description which follows, and in part will be obvious from the description, or may be learned by the practice of the teachings herein. Features and advantages of the disclosure may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. Features of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the disclosure as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] In order to describe the manner in which the above-recited and other features of the disclosure can be obtained, a more particular description will be rendered by reference to specific implementations thereof which are illustrated in the appended drawings. For better understanding, the like elements have been designated by like reference numbers throughout the various accompanying figures. While some of the drawings may be schematic or exaggerated representations of concepts, at least some of the drawings may be drawn to scale. Understanding that the drawings depict some example implementations, the implementations will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0009] FIG. 1 illustrates an example spreadsheet where a user is trying to author a new formula.

[0010] FIG. 2 illustrates an example environment for automatically recommending formulas for a spreadsheet in accordance with implementations of the present disclosure.

[0011] FIG. 3A illustrates an example of a high confidence of similar sheets with the same sheet names between a pair of spreadsheets in accordance with implementations of the present disclosure.

[0012] FIG. 3B illustrates an example of a low confidence of similar sheets with the same sheet names between a pair of spreadsheets in accordance with implementations of the present disclosure.

[0013] FIG. 3C illustrates an example of a low confidence of similar sheets with different sheet names between a pair of spreadsheets in accordance with implementations of the present disclosure.

[0014] FIG. 4 illustrates an example of a machine learning model architecture for spreadsheet representation in accordance with implementations of the present disclosure.

[0015] FIG. 5 illustrates example view windows for regions of a spreadsheet in accordance with implementations of the present disclosure.

[0016] FIG. 6 illustrates an example target sheet of a spreadsheet where a user is trying to author a new formula and an example reference sheet similar to the target sheet in accordance with implementations of the present disclosure.

[0017] FIG. 7 illustrates an example method for training machine learning models to identify similar spreadsheets or spreadsheet regions in accordance with implementations of the present disclosure.

[0018] FIG. 8 illustrates an example method for automatically recommending formulas for a spreadsheet in accordance with implementations of the present disclosure.

[0019] FIG. 9 illustrates components that may be included within a computer system.

DETAILED DESCRIPTION

[0020] This disclosure generally relates to formulas in spreadsheets. Spreadsheets are an electronic document in which data is arranged in rows and columns of a grid and can be manipulated and used in calculations. Spreadsheets also store tabular data in a row-and-column format. Spreadsheets are two dimensional but allow flexible organization of data (table content), metadata (captions, hierarchical headers, etc.), and free-form texts organized in ad-hoc ways. Unlike database tables or CSV/JSON tables, where each cell is just a string and a traditional database column may be stored in the horizontal direction, spreadsheets come with rich non-text styles (e.g., font, color, borders, size, etc.), which are meant to enhance readability for users and offer clues for users.

[0021] Spreadsheets, such as those in MICROSOFT EXCEL, are commonly recognized as popular end-user programming tools to manipulate tabular data. The intuitive spreadsheet interface combines the power of formula-based computation, with the ability to visually inspect tables. Spreadsheets are widely used by billions of non-technical users (e.g., the average enterprise users), who are typically neither database experts nor professional programmers.

[0022] Despite the popularity of spreadsheets, authoring formulas in spreadsheets remains to be a key pain point for non-technical users. Spreadsheets blend both data and code/formulas, in the same two-dimensional grid, where formulas are programmed at the granularity of individual cells (e.g., the formula in a cell can often be different from the formula in the neighboring cell). In order to create any non-trivial formula (beyond average (AVG()) and (SUM()) in spreadsheets), users first need to look for relevant functions, and understand the function syntax from reading documentations, which is non-trivial and close to actual programming that typically is performed by developers.

[0023] FIG. 1 illustrates an example spreadsheet **100** where a user is trying to author a new formula in a cell D41 with the intent to count the number of occurrences of the value displayed to the left ("Brown", in cell C41), within the column from the cell C7 to C37. The ground truth formula is therefore "=COUNTIF(C7:C37, C41)", which however is non-trivial for users to write, because COUNTIF() and its syntax is not something that an average non-technical user is familiar with. While the example in FIG. 1 illustrates just one function (COUNTIF), typically users in practice are trying to create formulas that have multiple functions with multiple parameters, each of which needs to be programmed correctly, and then stringed together appropriately, making the task of authoring formulas in spreadsheets clearly challenging.

[0024] Existing solutions try to predict formulas in a user-specified target cell based on surrounding Natural Language (NL) context. If the row label or column header of the target cell is "Total" or "Aggregate," the existing solutions use the row labels or column headers to determine that the desired formula will likely involve the function SUM() for the corresponding row or column. However, formulas with multiple parameters (e.g., COUNTIF() with two parameters) is difficult to infer from the natural language context alone, because even as humans, we may not be able to correctly guess this COUNTIF() formula by looking at the surrounding NL-context alone (without knowing the actual intent of the user for a formula in Cell D41). Existing solutions tend to have low accuracy in predicting formulas and are limited to predicting short formulas.

[0025] The present disclosure provides methods and systems that accurately predicts formulas that users want to author in a target spreadsheet cell by learning and adapting formulas that already exist in similar spreadsheets. The present disclosure includes a number of practical applications that provide benefits and/or solve problems associated with formulas in spreadsheets. Examples of these applications and benefits are discussed in further detail below.

[0026] The methods and systems represent spreadsheets and spreadsheets regions in a similar manner as images, where each cell is represented similar to a pixel. The methods and systems develop deep learning machine learning models to learn suitable dense vector representations for both spreadsheets and spreadsheet regions. Using the vector representation of spreadsheets and spreadsheet regions, the methods and systems can quickly find both similar spreadsheets and similar spreadsheet regions. In some implementations, the methods and systems use approximate

nearest neighbor (ANN) searches for vectors.

[0027] The methods and systems identify similar reference spreadsheets to a target spreadsheet. A target spreadsheet is a spreadsheet currently edited by users. A reference spreadsheet is an existing spreadsheet. In some implementations, the reference spreadsheets are existing spreadsheets within an organization. The methods and systems use a trained machine learning model to identify the similar reference spreadsheets.

[0028] One example use of the methods and systems of the present disclosure is automatically identifying similar spreadsheets in a corpus of spreadsheets (e.g., a collection of spreadsheets for an entity) to a target spreadsheet and performing downstream tasks on the similar spreadsheets. One example task is automatically marking similar spreadsheets with a same category or label as a target spreadsheet. For example, a user identifies a target spreadsheet as confidential, the methods and systems automatically identify other similar spreadsheets in the corpus and automatically apply a confidential label to the identified similar spreadsheets. Instead of the user manually labeling or identifying other spreadsheets as confidential, the methods and systems automatically identify similar spreadsheets and automatically indicate that the similar spreadsheets are confidential. Another example task is setting user access permissions for accessing a spreadsheet. For example, an accounting group of a company (e.g., a subset of users) are assigned access permissions to the financial report spreadsheets of the company. A user may access a financial report spreadsheet and the methods and systems may automatically identify similar spreadsheets in the company's spreadsheets to the financial report spreadsheet and automatically set the user access permissions to the similar spreadsheets to the subset of users in the financial group of the company. The methods and systems may aid users in quickly identifying spreadsheets the users have permission to access.

[0029] The methods and systems may also identify similar reference regions to a target cell. A cell is the intersection of a row and a column in a spreadsheet (where a row and column meet). Columns are identified by letters (A, B, C, etc.) and rows are identified by numbers (1, 2, 3, etc.). Each cell has its own name or cell address based on its column and row. A region is one or more cells within a spreadsheet. A target region is a region (one or more cells) within a target spreadsheet. A reference region is a region (one or more cells) within an existing spreadsheet. A target cell is a cell within a spreadsheet in which a user wants to create a formula. A formula is a user-programmed expression in a cell, which consists of functions and parameters to these functions, which are often cell-locations, such as "B5," that refer to data in other cells within the spreadsheet. The methods and systems use a trained machine learning model to identify similar reference regions to a target cell.

[0030] The methods and systems identify formulas within the similar reference cells from the similar spreadsheets and automatically adapt the formulas to the target cell as a predicted formula for the target cell. The methods and systems use a trained machine learning model to identify and adapt the formulas from the similar reference cells to the predicted formula for the target cell.

[0031] The methods and systems provide a formula recommendation with the predicted formula to the user. In some implementations, the methods and systems automatically insert the predicted formula in the target cell. The user may accept the predicted formula or modify the predicted formula.

[0032] One technical advantage of the systems and methods of the present disclosure is improved quality in the predicted formula. The methods and systems use more training examples to train the machine learning models, resulting in a more accurate machine learning model and a machine learning model that provides better precision and recall. Another technical advantage of the systems and methods of the present disclosure is an efficiency optimization. The systems and methods of the present disclosure produce an embedding of a vector representation of the spreadsheet or spreadsheet region and use the embeddings to identify similar spreadsheets or spreadsheet regions quickly and efficiently.

[0033] Referring now to FIG. 2, illustrated is an example environment **200** for automatically

recommending formulas for a spreadsheet. The environment **200** includes an offline phase **202** for training a plurality of machine learning models (e.g., the machine learning model **206**, the machine learning model **208**, and the machine learning model **210**) identifying similar spreadsheets and similar regions of spreadsheets. The offline phase **202** automatically trains the machine learning model **206** to generate training examples of similar vs. dissimilar spreadsheets, from a plurality of spreadsheets (e.g., spreadsheets **10a**, **10b**, **10c** up to n, where n is a positive integer), using weak supervision. In some implementations, the plurality of spreadsheets (e.g., spreadsheets **10a**, **10b**, **10c**) are obtained from publicly available databases. One example use case includes crawling the internet and using 160,000 spreadsheets as the plurality of spreadsheets. Any number of spreadsheets may be used by the machine learning model **206** to automatically generate the training examples using weak supervision.

[0034] In some implementations, the machine learning model **206** uses the sheet names of the spreadsheets to determine whether the spreadsheets are similar or dis-similar to each other. Each spreadsheet file typically contains multiple sheets, where each sheet has a name and each sheet is a two-dimension grid, similar to a traditional table. The machine learning model **206** compares the sheet names of a pair of spreadsheet files to determine whether the pair of spreadsheet files are similar or dis-similar to each other.

[0035] For example, when two spreadsheet files (F and F') contain multiple sheets (s and s'), denoted as F=(s.sub.1, s.sub.2, . . . s.sub.n) and F'=(s'.sub.1, s'.sub.2, . . . s'.sub.n), and all the names of these sheets match exactly 1-to-1 with each other, or name(s.sub.i)=name(s'.sub.i), $\forall i \in [n]$, then these two files (F and F'), as well as their corresponding sheets (s.sub.i and s'.sub.i), are likely to be similar. However, if the sheet names share identical names (a match occurs between the sheet names), it does not mean that the two spreadsheets (F and F') must be similar. For example, if the sheets have a default name (e.g., sheet1) or a common name, there may not be enough information based on name of the sheets for the machine learning model **206** to use in determining whether the spreadsheets are similar or dis-similar. If a match does not occur between the names of the sheets, the corresponding sheets are likely to be dis-similar.

[0036] In some implementations, the machine learning model **206** uses a weak supervision hypothesis test to model the probabilities of seeing different sequences of sheet-names (common vs. rare names, long vs. short sequences) to use in determining whether sheet names are common sheet names or rare sheet names. The hypothesis test includes a null hypothesis H.sub.0 that states that two given spreadsheets F and F' are not similar (a default position that should apply in the majority of cases), and for the sheet-names of (s.sub.1, s.sub.2, . . . s.sub.n) and (s'.sub.1, s'.sub.2, . . . s'.sub.n) to match is a just coincidence. The hypothesis test explicitly models the probabilistic process of drawing two sequences of sheets (s.sub.1, s.sub.2, . . . s.sub.n) and (s'.sub.1, s'.sub.2, . . . s'.sub.n) from a universe of sheet-names. If the machine learning model **206** determines the probability of two given sequences of names to collide is exceedingly small (e.g., smaller than typical a thresholds, such as, **0.05**), the machine learning model **206** rejects the null hypothesis (H.sub.0) and concludes that this is not a coincidence, the two files F and F', as well as their sheets, are likely similar.

[0037] An example equation that the machine learning model **206** uses in determining whether sheet names are common sheet names or less common sheet names is illustrated below in equation (1):

$$[00001] \ p_i = \frac{\text{freq}_U(\text{name}(s_i))}{|\text{Math. } U \cdot \text{Math.}|} \quad (1)$$

where $p_i = \text{Pr}(\text{name}(s_{\text{sub}.i}))$ is the probability of encountering the sheet-name name(s.sub.i), when a random spreadsheet is drawn from all sheets in the universe, U is the universe of all spreadsheets, $\text{freq}_U(\text{name}(s_{\text{sub}.i}))$ is the frequency of name(s.sub.i) in all spreadsheets in U, and |U| is the total number of spreadsheets.

[0038] Given two sequences of sheets (s.sub.1, s.sub.2, . . . s.sub.n) and (s'.sub.1, s'.sub.2, . . .

$s'.sub.n$), the chance of seeing the sheet-names in a sequence to match exactly with the first may be calculated by the machine learning model **206** as $\prod_{i \in [n]} p_{sub.i}$ (because each time a spreadsheet $s'.sub.i$ is drawn, the probability that its name matches the name from the first sequence, $name(s.sub.1)$, is $p_{sub.i}$, $\forall i \in [n]$). Under the null-hypothesis, if the machine learning model **206** finds the probability of two sequences of sheets sharing identical names, $\prod_{i \in [n]} p_{sub.i}$, to be exceedingly small, e.g., smaller than the typical alpha of 0.05, the machine learning model **206** rejects $H.sub.0$ and concludes that $(s.sub.i, s'.sub.i)$ are similar sheets, for all $i \in [n]$. [0039] In some implementations, the machine learning model **206** uses the hypothesis test as weak-supervision to automatically generate positive and negative training pairs for similar and dis-similar sheets. To generate positive examples for similar sheets, the machine learning model **206** checks pairs of spreadsheet files (F, F') , whose sequences of sheets $(s.sub.1, s.sub.2, \dots s.sub.n)$ and $(s'.sub.1, s'.sub.2, \dots s'.sub.n)$ have identical names. The machine learning model **206** marks all $(s.sub.1, s'.sub.i)$ $\forall i \in [n]$ as positive examples (similar sheets), if $\prod_{i \in [n]} p_{sub.i} \leq \alpha$, (e.g., $\alpha=0.05$), and rejects $H.sub.0$.

[0040] FIG. 3A illustrates an example of a high confidence of similar sheets between a pair of spreadsheets (the spreadsheet **302** and the spreadsheet **304**) with the same sheet names. The spreadsheet **302** includes a sheet **306** with a sheet name “Instructions,” a sheet **308** with a sheet name “WorkshopDetails,” a sheet **310** with a sheet name “Workshop_Outputs,” a sheet **312** with a sheet name “Report Summary,” a sheet **314** with a sheet name “Report Theme,” and a sheet **316** with a sheet name “DataByBranch.” The spreadsheet **304** includes a sheet **318** with a sheet name “Instructions,” a sheet **320** with a sheet name “WorkshopDetails,” a sheet **322** with a sheet name “Workshop_Outputs,” a sheet **324** with a sheet name “Report Summary,” a sheet **326** with a sheet name “Report Theme,” and a sheet **328** with a sheet name “DataByBranch.”

[0041] In the illustrated example, the sequence of sheets (e.g., the sheets **306, 308, 310, 314, 316** and the sheets **318, 320, 322, 324, 326, 328**) have identical sheet names for the pair of spreadsheets **302, 304** in the same order. For example, the sheet names of the first sheets (e.g., the sheet **306** and the sheet **318**) are the same (e.g., “Instructions”), the sheet names of the second sheets (e.g., the sheet **308** and the sheet **320**) are the same (e.g., “WorkshopDetails”), the sheet names of the third sheets (e.g., the sheet **310** and the sheet **322**) are the same (e.g., “Workshop_Outputs”), the sheet names of the fourth sheets (e.g., the sheet **312** and the sheet **324**) are the same (e.g., “Report Summary”), the sheet names of the fifth sheets (e.g., the sheet **314** and the sheet **326**) are the same (e.g., “Report Themes”), and the sheet names of the sixth sheets (e.g., the sheet **316** and the sheet **328**) are the same (e.g., “DatabyBranch”).

[0042] The machine learning model **206** (FIG. 2) uses equation (1) (FIG. 2) to automatically determine a probability of drawing a random sheet from a universe of sheets with the sheet names and uses the null-hypothesis test on the probability to determine whether the pair of spreadsheets **302, 304** are similar or dis-similar. The machine learning model **206** uses the equation (1) to identify whether the sheet names are common or uncommon. The machine learning **206** aggregates the probability of the sheet names and determines whether the aggregated probability is less than 0.05 (e.g., an uncommon sheet name) or greater than or equal to 0.05 (e.g., a common sheet name).

[0043] For example, the machine learning model **206** determines that the sheet name “Instructions” occurs 100 times for a universe of 100,000 spreadsheets, so the probability drawing a random sheet from the universe and seeing the name “Instructions” is 0.1% (100/100,000). Another example includes the machine learning model **206** determines that the sheet name “WorkshopDetails” occurs 10 times in total, so the probability is 0.0001% (10/100,000), which indicates that the sheet name “WorkshopDetails” is less common as compared to the sheet name “Instructions.” The machine learning model determines that the aggregated probability of the sheet names is less than 0.05, rejects the null-hypothesis ($H.sub.0$), and concludes that the two sequence of sheets from the pair of spreadsheets **302, 304** are similar. The machine learning model **206** uses the pair of spreadsheets (the spreadsheet **302** and the spreadsheet **304**) as a positive example of a training pair

of spreadsheets (e.g., similar spreadsheets).

[0044] Referring to FIG. 2, to generate negative examples for similar sheets, the machine learning model **206** samples two random sheets as negative examples, because the chance of collision (the two sheets being similar) in a large spreadsheet corpus is small. The machine learning model **206** samples pairs of spreadsheet files (F, F'), and only if the two sets of sheets (s.sub.1, s.sub.2, . . . s.sub.n) and (s'.sub.1, s'.sub.2, . . . s'.sub.n) do not share any common sheet-names, the machine learning model **206** uses pairs of (s.sub.i, s'.sub.j) as negative examples (dis-similar sheets).

[0045] In some implementations, the machine learning model **206** uses pairs of spreadsheets as a negative example (dis-similar sheets) if the sheets have a default name (e.g., sheet1) or a common name that are the same. The machine learning model **206** may not have enough information to reject the null hypothesis.

[0046] FIG. 3B illustrates an example of a low confidence of similar sheets between a pair of spreadsheets (the spreadsheet **330** and the spreadsheet **332**) with the same sheet names. The spreadsheet **330** includes a sheet **334** with a sheet name "Sheet1" and the spreadsheet **332** includes a sheet **336** with a sheet name "Sheet1." The machine learning model **206** (FIG. 2) uses equation (1) (FIG. 2) to automatically determine a probability of drawing a random sheet from a universe of sheets with the sheet name "Sheet1" and uses the probability to identify whether the sheet name "Sheet1" is a common sheet name or an uncommon sheet name.

[0047] For example, the machine learning model **206** determines the sheet name "Sheet1" occurs 15,000 times out of a total of 100,000 spreadsheets and the probability value for the sheet name "Sheet1" is 15,000/100,000 (0.15%). The machine learning model **206** compares the probability value of the sheet name (e.g., 0.15) and determines that 0.15 is greater than 0.05 and keeps the null-hypothesis (H.sub.0) (e.g., the machine learning model **206** has low confidence that the spreadsheets **330**, **332** are similar). The machine learning model **206** concludes that the two sequence of sheets from the pair of spreadsheets **330**, **332** are dis-similar spreadsheets and uses the pair of spreadsheets (the spreadsheet **330** and the spreadsheet **332**) as an example of a negative training pair of spreadsheets (e.g., dis-similar spreadsheets).

[0048] FIG. 3C illustrates an example of a low confidence of similar sheets between a pair of spreadsheets (the spreadsheet **338** and the spreadsheet **340**) with different sheet names. The spreadsheet **338** includes a sheet name "Option B" and the spreadsheet **340** includes a sheet name "SRAC." The pair of spreadsheets (the spreadsheet **338** and the spreadsheet **340**) have different sheet names. The machine learning model **206** uses weak supervision and determines that the pair of spreadsheets (the spreadsheet **338** and the spreadsheet **340**) are dis-similar to one another in response to determining that the sheet name "Option B" and the sheet name "SRAC" are different. The machine learning model **206** identifies the pair of spreadsheets (the spreadsheet **338** and the spreadsheet **340**) as an example of a negative training pair of spreadsheets (e.g., dis-similar spreadsheets).

[0049] Referring back to FIG. 2, to generate positive example of similar regions of spreadsheet files, the machine learning model **206** takes a pair of similar sheets (s.sub.i, s'.sub.i), and checks for all formulas on s.sub.i and s'.sub.i. For example, if formula $f \in s.sub.i$ and $f' \in s'.sub.i$, where the locations of f and f' are identical, or $Loc(f) = Loc(f')$ (e.g., one is in B59 of s.sub.i while the other is also in B59 of s'.sub.i), and their formula-expressions are also identical, or $f = f'$ (e.g., both are SUM(A12:B40)), the machine learning model **206** has a high confidence that the regions surrounding Loc (f) and Loc (f') are similar-regions, which the machine learning model **206** uses as positive examples.

[0050] To generate negative examples for similar-regions (regions that are dis-similar), the machine learning model **206** takes the positive examples (Loc (f), Loc (f')) determined for the similar regions, and shifts one of the locations Loc (f') until the location hits a different formula g with $g \neq f$, which is when the machine learning model **206** stops and marks (Loc (f), Loc (g)) as a negative example for similar-region. The weak supervision based on the hypothesis tests when applied to a

spreadsheet corpus automatically generates large amounts of positive and negative examples for similar sheets and/or similar regions of spreadsheets.

[0051] The machine learning model **206** outputs training triples **12** that include positive and negative examples that correspond to similar and dis-similar sheets and/or regions of spreadsheet files. In some implementations, the training triples **12** include triplet examples (A, P, N), where A is an anchor (a reference spreadsheet), P is a positive example to the anchor (a similar spreadsheet), and N is a negative example to the anchor (a dis-similar spreadsheet). For example, the machine learning model **206** uses the below equation (2) to generate the training triples **12**.

$$[00002] \quad l_{\text{triplet}} = \max(\text{.Math.}_{A-P}^2 - \text{.Math.}_{A-N}^2 + m, 0) \quad (2)$$

where (A, P, N) are three spreadsheet-regions, with (A, P) being a pair of automatically generated positive examples using weak supervision, (A, N) is a pair of negative examples generated using the weak supervision, and m refers to margin, which is a hyperparameter that encourages the machine learning model **206** to continue to push negative examples N further away from positive examples P, relative to anchor A, until there is a safe margin to differentiate the two.

$\|\phi A - \phi P\|_{\text{sup.2}}$ measures the L.sub.2 distance between the representations for the two positive examples ϕA and ϕP , which after training is small, while $\|\phi A - \phi N\|_{\text{sup.2}}$ is the L.sub.2 distance between two negative examples, which is large.

[0052] In some implementations, the machine learning model **206** uses the below Algorithm (1) for the training.

TABLE-US-00001 Algorithm 1: Offline training Input: A large corpus of spreadsheets S Output: Coarse-grained model M.sub.c, Fine-grained model M.sub.f
1 Initialize M.sub.c, M.sub.f with random parameters.
2 Pr.sub.s.sub.P, Pr.sub.s.sub.N, Pr.sub.r.sub.P, Pr.sub.r.sub.N = TrainDataGen(S)
3 for each episode $\in [1, T]$ do
4 | ($\phi_{\text{sub.c.sub.A}}, \phi_{\text{sub.c.sub.P}}, \phi_{\text{sub.c.sub.N}}$) = SHSample(Pair.sub.s.sub.P, Pair.sub.s.sub.N.sub., Mc)
5 | ($\phi_{\text{sub.f.sub.A}}, \phi_{\text{sub.f.sub.P}}, \phi_{\text{sub.f.sub.N}}$) = SHSample(Pair.sub.r.sub.P, Pair.sub.r.sub.N.sub., Mf)
6 | $l_{\text{sub.triplet.sub.c}} = \max(\|\phi_{\text{sub.c.sub.A}} - \phi_{\text{sub.c.sub.N}}\|_{\text{sup.2}} - \|\phi_{\text{sub.c.sub.A}} - \phi_{\text{sub.c.sub.P}}\|_{\text{sup.2}} + m, 0)$
7 | $l_{\text{sub.triplet.sub.f}} = \max(\|\phi_{\text{sub.f.sub.A}} - \phi_{\text{sub.f.sub.N}}\|_{\text{sup.2}} - \|\phi_{\text{sub.f.sub.A}} - \phi_{\text{sub.f.sub.P}}\|_{\text{sup.2}} + m, 0)$
8 | Perform SGD training on M.sub.c with $l_{\text{sub.Triplet.sub.c}}$
9 | Perform SGD training on M.sub.f with $l_{\text{sub.Triplet.sub.f}}$
10 end

[0053] During each step of the training described in Algorithm (1), the machine learning model **206**, continuously selects semi-hard triples (A, P, N) that are the most informative for models to learn, based on the current state of the models. Specifically, the machine learning model **206** samples triples (A, P, N) whose loss satisfies $0 < l_{\text{sub.triplet}}(A, P, N) < m$, which ensures that the selected triple (A, P, N) is neither too hard (with loss greater than m), nor too easy (with loss equals 0).

[0054] The training triples **12** are used to train the spreadsheet representative models (the machine learning model **208** and the machine learning model **210**) so that when the representative models are given a new pair of sheets (s, s') or regions (Loc(c), Loc(c')) of spreadsheets, the spreadsheet representative models may quickly and accurately compute a similarity between the sheets and/or regions. In some implementations, the machine learning model **208** is used to identify whether the pair of sheets are similar or dis-similar and the machine learning model **210** is used to identify whether the pair of regions are similar or dis-similar.

[0055] The spreadsheet representation-models (the machine learning model **206** and the machine learning model **210**) are universal and applicable to all types of spreadsheets, and therefore, the learning steps happens once during the offline phase **202**.

[0056] In some implementations, indexes are generated in the offline phase **202** to index the

spreadsheets S in an organization. Given a new corpus of spreadsheets S in an organization, the representation models (the machine learning models **208**, **210** learned on U) to generate vector representations for each spreadsheet S . S are provided to the indexes. For example, a similar spreadsheet index is created with similar spreadsheets that is used to quickly find similar spreadsheets in step **214**. Another example includes a similar region index is created with similar spreadsheet regions that can be used to quickly find similar regions in steps **216**, **218**. In some implementations, the indexes are approximate nearest neighbor (ANN) indexes. The indexes aid in efficient and accurate identification of similar spreadsheets and/or spreadsheet regions in the online phase **204**.

[0057] The environment **200** also includes an online phase **204** that a user **212** accesses when using a target sheet **14** (a spreadsheet currently being edited by the user **212**). In some implementations, the online phase **204** is triggered when the user **212** opens the target sheet **14**, for example, using an application (e.g., a spreadsheet application) on a device of the user **212** or a browser on the device of the user **212**.

[0058] The online phase **204** includes a search for reference sheet step **214**. The machine learning model **208** (trained in the offline phase **202**) searches to identify one or more reference sheets, denoted by $SR \in S$ (S is the spreadsheet corpus that the user **212** is able to access), that are similar to the target sheet **14** (ST). The machine learning model **208** uses a similar sheet primitive to identify the one or more reference sheets. In some implementations, the search for reference sheet step **214** occurs in response to the user **212** opening the target sheet **14**. The search for reference sheet step **214** may occur prior to the user **212** starting a formula in a target cell of the target sheet **14**. For example, the indexes are accessed in response to the user **212** opening the target sheet **14** to automatically identify similar sheets to the target sheet **14**.

[0059] The online phase **204** also includes a search for reference formula step **216** by similar region. The machine learning model **210** searches within each reference sheet SR to identify a reference cell $C.sub.R \in S.sub.R$, whose local spreadsheet regions look similar to the local region around the target-cell $C.sub.T$, using a similar region primitive. If $C.sub.R$ contains a formula $F.sub.R$, the reference formula is identified by the machine learning model **210** as a likely candidate for a predicted formula **16**. In some implementations, the machine learning model **210** identifies a plurality of reference formulas that are likely candidates for the predicted formula **16** and places the plurality of reference formulas in a ranked order. The machine learning model **210** may take a highest ranked reference formula as the predicted formula.

[0060] In some implementations, the search for reference formula step **216** automatically occurs in response to the machine learning model **210** receiving an indication that the user **212** is creating a formula in a target cell. One example indication that the user **212** is creating a formula includes the user **212** typing an equal sign in a target cell. Another example indication that the user **212** is creating a formula includes the user **212** typing a partial formula in a target cell. Another example indication that the user **212** is creating a formula includes the user **212** selecting a create formula option.

[0061] The online phase **204** also includes a search parameter cell step **218** by similar region. The machine learning model **210** uses the similar region primitive to adapt the reference formula ($F.sub.R$), into the local context of the target-sheet (ST), by changing the parameter cells of the reference formula ($F.sub.R$). The machine learning model **210** uses the similar-region primitive to learn-to-adapt these parameters from SR to ST and output a predicted formula **16** with specific content from the target sheet **14**.

[0062] The predicted formula **16** is automatically displayed in the target cell. In some implementations, the predicted formula **16** is displayed in response to the user **212** typing the equal sign prior to the user **212** typing any additional information for the formula. In some implementations, the predicted formula **16** is a partial recommendation for the formula. If the user **212** accepts the predicted formula **16**, the machine learning model **210** provides a next portion of

the predicted formula **16**. For example, the user accepts the predicted formula **16** by keeping the predicted formula **16** in the target cell. The user **212** may accept the predicted formula **16** or make any modifications or changes to the predicted formula **16**. For example, the user **212** may override the predicted formula **16** by typing in the desired parameters and functions in the target cell. [0063] In some implementations, the user **212** may continue to type a formula in the target cell and the machine learning model **210** updates the predicted formula **16** in response to the changes in the formula as the user **212** is typing. For example, the machine learning model **210** refines the predicted formula **16** based on the prefix the user **212** is typing for the formula. The machine learning model **210** may continue to refine the predicted formula **16** until the user **212** finishes typing the formula in the target cell.

[0064] In some implementations, one or more computing devices (e.g., servers and/or devices) are used to perform the processing of the environment **200**. The one or more computing devices may include, but are not limited to, server devices, cloud virtual machines, personal computers, a mobile device, such as, a mobile telephone, a smartphone, a PDA, a tablet, or a laptop, and/or a non-mobile device. The features and functionalities discussed herein in connection with the various systems may be implemented on one computing device or across multiple computing devices. For example, the machine learning models **206**, **208**, **210**, the spreadsheet corpus, and the spreadsheet application are implemented wholly on a computing device. Another example includes the machine learning models **206**, **208**, **210**, the spreadsheet corpus, and/or the spreadsheet application implemented wholly on a computing device. Another example includes one or more subcomponents of each of the machine learning models **206**, **208**, **210**, the spreadsheet corpus, and the spreadsheet application implemented across multiple computing devices. Moreover, in some implementations, one or more subcomponent of the machine learning models **206**, **208**, **210**, the spreadsheet corpus, and/or the spreadsheet application may be implemented are processed on different server devices of the same or different cloud computing networks.

[0065] In some implementations, each of the components of the environment **200** is in communication with each other using any suitable communication technologies. In addition, while the components of the environment **200** are shown to be separate, any of the components or subcomponents may be combined into fewer components, such as into a single component, or divided into more components as may serve a particular implementation. In some implementations, the components of the environment **200** include hardware, software, or both. For example, the components of the environment **200** may include one or more instructions stored on a computer-readable storage medium and executable by processors of one or more computing devices. When executed by the one or more processors, the computer-executable instructions of one or more computing devices can perform one or more methods described herein. In some implementations, the components of the environment **200** include hardware, such as a special purpose processing device to perform a certain function or group of functions. In some implementations, the components of the environment **200** include a combination of computer-executable instructions and hardware.

[0066] FIG. **4** illustrates an example of a machine learning model architecture **400** for spreadsheet representation to classify spreadsheets and/or spreadsheet regions as similar or dis-similar. The machine learning model architecture **400** is used by the machine learning model **206** (FIG. **2**). In some implementations, the machine learning model **206** is a classification machine learning model. The machine learning model architecture **400** includes input features layers **402**, dimension reduction layers **404**, feature extraction layers **406**, and output layers **408**.

[0067] The input feature layers **402** represent the spreadsheets and the spreadsheet regions as input feature vectors. The input feature vectors are provided to deep neural networks (DNN) machine learning models for representation learning. The input feature layers **402** converts the two-dimensional grids consisting of cells of the spreadsheets or the spreadsheet regions into the input feature vectors. Spreadsheet cells contain content and rich style features (e.g., color, font, font-size,

borders, etc.). The input feature layers **402** use content features and style features for each spreadsheet cell to convert the cell features **18** into the input feature vectors (the semantic feature vector **20**, the syntactic feature vector **22**, the style feature vector **24**).

[0068] The content features ($\gamma_{\text{sup.c}}$) represent the content in a cell. The content features include semantic and syntactic features. Semantic features are components of a concept associated with a lexical item. The semantic features of the content in the cell help identify a word's meaning. In some implementations, a pre-trained natural language embedding is applied to the content in a cell to produce a vector representation (the semantic feature vector **20**) of each cell value. Semantically similar content has embeddings that are close in the semantic feature vector **20** and semantically dissimilar content has embeddings that are further away from each other in the semantic feature vector **20**. For example, “USA” and “Canada,” are close by in the feature space of the semantic feature vector **20**.

[0069] The syntactic feature vector **22** represents the data type of each cell (e.g., numeric, text, empty, etc.) as categorical features. The syntactic feature vector **22** also represents the syntactic-patterns of values (e.g., “DDDD-DD-DD” for “2020-01-02”) as features. Example syntactic features include numbers in the cell, patterns in the cell, and/or a format of the content in the cell.

[0070] The style feature vector **24** represents styles encoded in the spreadsheet cells as a syntactic feature vector **22** ($\gamma_{\text{sup.s}}$). Example style features include background color, font color, font style (bold, italic, etc.), font size, cell size (height and width), boundaries of the cell, and/or a cell without boundaries. The final feature vector ($\gamma(C)$) for each cell (C) of the spreadsheet is a concatenation of $\gamma_{\text{sup.c}}(C)$ and $\gamma_{\text{sup.s}}(C)$.

[0071] Spreadsheets may not have table boundaries. For example, there may be multiple irregularly shaped tables on a sheet. In some implementations, a view window **28** with rows ($n_{\text{sub.r}}$) and columns ($n_{\text{sub.c}}$) (where n is a positive integer) is used to represent a surrounding region of a cell (C). One example includes $n_{\text{sub.r}}$ is 100 rows and $n_{\text{sub.c}}$ is 10 columns. The view window **28** may move around in a spreadsheet to represent different regions of the spreadsheet. In some implementations, the view window **28** is centered at the location of a cell (C) in the sheet and is denoted as $V(C)$. In some implementations, the view window **28** represents an entire spreadsheet by starting at a top corner of the spreadsheet.

[0072] FIG. 5 illustrates example view windows **28.sub.1**, **28.sub.2** for regions of a spreadsheet **500**. The view window **28.sub.1** represents the surrounding region of a cell **502** (the cell A120). For example, the view window **28.sub.1** is the ($n_{\text{sub.r}} \times n_{\text{sub.c}}$) box (100 rows by 10 columns) centered around the cell A120 (the view window $V(A120)$). The view window **28.sub.2** represents an entire spreadsheet using a ($n_{\text{sub.r}} \times n_{\text{sub.c}}$) box (100 rows by 10 columns) starting from the top-left corner of a spreadsheet.

[0073] Referring back to FIG. 4, the input vector **26** (ϕv) for a view window V , representing a spreadsheet or a spreadsheet region, is the concatenation of the cell-features for all cells in the view window ($\{\gamma(C) | \forall C \in V\}$). In some implementations, the cell-features are concatenated using the row-major order of the two-dimension region to produce the input vector **26**.

[0074] In some implementations, the input vector **26** is a concatenation of different types of features for each cell resulting in hundreds of dimensions. The dimension reduction layers **404** use shared weights for each cell to distill the features from the input vector **26** to use for the spreadsheet representations. In some implementations, the dimension reduction layers **404** are implemented as Multi-Layer Perception (MLP) layers.

[0075] The input vector **26** for each cell is provided to the machine learning model **208** and the machine learning model **210** in the feature extraction layers **406**. In some implementations, the features identified in the dimension reduction layers **404** are provided in the input vector **26** to the machine learning model **208** and the machine learning model **210**.

[0076] The machine learning model **208** is trained to identify similar sheets in spreadsheets. In some implementations, the machine learning model **208** is a convolutional neural network that is

insensitive to shifts of rows and columns and blurs boundaries between cells to create a representation of the spreadsheets to determine whether the spreadsheets are similar.

[0077] The machine learning model **210** is trained to identify similar regions in spreadsheets. In some implementations, the machine learning model **210** has fully connected networks to represent the cell boundaries that the machine learning model **210** uses to identify precise similar regions.

[0078] The output of the features extraction layers **406** are dense vector representations of coarse grained views **30** and fine grained views **32** of a spreadsheet region. The output layers **408** perform L.sub.2 normalization **34**, **36** on the coarse grained views **30** and the fine grained views **32**. The resulting dense vectors are used by the machine learning model **208** and the machine learning model **210** to directly compute similarities between two spreadsheets or spreadsheet regions. For example, the vectors **38** are used by the machine learning model **208** to compute similarities between two spreadsheets and the vectors **40** are used by the machine learning model **210** to compute similarities between two spreadsheet regions. In some implementations, the output layers **408** uses the dense vectors produced from the output layers **408** to compute the training triples **12** (FIG. 2). The output of the machine learning model **208** and the machine learning model **210** is whether the spreadsheets and/or spreadsheet regions are similar or dis-similar.

[0079] FIG. 6 illustrates an example target sheet **14** of a spreadsheet where a user (e.g., the user **212** (FIG. 2)) is trying to author a new formula in the target cell **52** (D41) and an example reference sheet **42** that is similar to the target sheet **14**. The user **212** opens the target sheet **14** and the machine learning model **208** performs the search for reference sheets step **214** (FIG. 2) and identifies the reference sheet **42** as similar to the target sheet **14**. The machine learning model **208** searches a spreadsheet corpus, for example, a spreadsheet corpus of an organization of the user **212**, to find reference-sheets that look similar to the target sheet **14**.

[0080] In some implementations, an index is used to quickly identify a similar reference sheet **42**. In some implementations, a plurality of similar reference sheets is identified by the machine learning model **208** and the machine learning model **208** selects the reference sheet **42** (e.g., a top ranked reference sheet) from the plurality of similar reference sheets.

[0081] The machine learning model **210** (FIG. 2) performs the search for reference-formula step **216** (FIG. 2) in response to determining the user **212** is authoring a formula (e.g., an equal sign is typed or other indication a formula is being created). The machine learning model **210** identifies the reference formula **46** in the reference cell **44** (D354) in response to determining that the region **48** surrounding the reference cell **44** is a similar region to the region **50** surrounding the target cell **52**. The machine learning model **210** uses the formula “=COUNTIF(C6:C350, C354)” in the reference cell **44** as the reference formula **46**.

[0082] The machine learning model **210** (FIG. 2) performs the search parameter cells step **218** (FIG. 2) and adapts parameters of the reference formula **46** to the context of the target cell **52** to generate the predicted formula **16**. The machine learning model **210** uses a formula template for the reference formula **46** and fills in the parameters of the reference formula **46** using the local context of the target cell **52** to generate the predicted formula **16**. The original parameters C6, C350 and C354 in the reference formula **46** are likely to be incorrect for the target cell **52**. The machine learning model **210** identifies similar regions in the target sheet **14** (e.g., the regions around C7, C37, and C41 of the target sheet **14**) that are similar to regions in the reference sheet **42** (e.g., the regions around C6, C37, and C41 of the reference sheet **42**) and updates the parameters of the predicted formula **16** using the similar regions of the target sheet **14** (C7, C37, and C41).

[0083] The predicted formula **16** is automatically presented in the target cell **52** to the user **212**. The user **212** may accept the predicted formula **16**. In some implementations, the user **212** may continue typing, changing, or modifying the predicted formula **16**.

[0084] FIG. 7 illustrates an example method **700** for training machine learning models to identify similar spreadsheets or spreadsheet regions. The actions of the method **700** are discussed below with reference to FIGS. 2-6.

[0085] At **702**, the method **700** includes automatically generating, by a machine learning model, training data with positive examples of similar spreadsheets and negative examples of dis-similar spreadsheets. The machine learning model **206** accesses a large number of spreadsheets (e.g., the spreadsheets **10a**, **10b**, **10c**) to automatically generate the training data (e.g., the training triples **12**). For example, the machine learning model **206** crawls over 100,000 spreadsheets to automatically generate the training data.

[0086] In some implementations, the machine learning model **206** uses the sheet names of the spreadsheets to determine whether the spreadsheets are similar or dis-similar to each other. Each spreadsheet file typically contains multiple sheets, where each sheet has a name and each sheet is a two-dimension grid, similar to a traditional table and the machine learning model **206** compares the sheet names of a pair of spreadsheet files to determine whether the pair of spreadsheet files are similar or dis-similar to each other. The machine learning model **206** uses a weak supervision hypothesis test to compare sheet names of pairs of spreadsheets in the training data and outputs a positive example (the spreadsheets are similar) in response to the sheet names matching and outputs a negative example (the spreadsheets are different) in response to the sheet names having different names. In some implementations, the machine learning model **206** uses a probability of finding the sheet names in the training data and outputs the positive example in response to determining the probability of the sheet names is less common (occurs infrequently) in the training data and outputs the negative example in response to determining the probability of the sheet names is common (occurs frequently) in the training data.

[0087] In some implementations, the machine learning **210** takes a pair of similar sheets and checks for all formulas on the similar sheets. If the formulas match, the machine learning model **210** has a high confidence that the regions surrounding the formulas are similar-regions and the machine learning model **206** uses the regions as positive training examples. To generate negative training examples, the machine learning model **206** uses the regions from the positive training examples, shifts the location of one of the sheets until a different formula is identified and marks the regions as a negative example.

[0088] In some implementations, the training data is represented as training triples **12** that include positive and negative examples that correspond to similar and dis-similar sheets and/or regions of spreadsheet files. In some implementations, the training triples **12** include triplet examples (A, P, N), where A is an anchor (a reference spreadsheet), P is a positive example to the anchor (a similar spreadsheet), and N is a negative example to the anchor (a dis-similar spreadsheet).

[0089] In some implementations, the training data is represented as a plurality of feature vectors with embeddings for each cell of the spreadsheets. For example, the plurality of feature vectors includes a semantic feature vector **20**, a syntactic feature vector **22**, and a style feature vector **24**. The plurality of feature vectors are aggregated together into an input vector **26** for each cell (e.g., the cell **502**) of the spreadsheet (e.g., the spreadsheet **500**) that represents a region around each cell (e.g., the window **28.sub.1**, **28.sub.2**) of the spreadsheet (e.g., the spreadsheet **500**).

[0090] At **704**, the method **700** includes using the training data, by a first machine learning model, to determine whether a pair of spreadsheets are similar. The training triples **12** are used to train the machine learning model **208** to identify similar spreadsheets so that when the machine learning model **208** is given a new pair of spreadsheets, the machine learning model **208** may quickly and accurately compute a similarity between the sheets. In some implementations, the machine learning model **208** uses the input vector **26** for each cell of a pair of spreadsheets to determine whether a pair of spreadsheets (e.g., the target sheet **14** and the reference sheet **42**) are similar.

[0091] At **706**, the method **700** includes using the training data, by a second machine learning model, to identify similar spreadsheet regions between the pair of spreadsheets. The training data (e.g., the training triples **12**) are used to train the machine learning model **210** to identify similar spreadsheet regions so that when the machine learning model **210** is given a new pair of spreadsheets, the machine learning model **210** may quickly and accurately identify similar

spreadsheet regions in the pair of spreadsheets. In some implementations, the machine learning model **210** uses the input vector **26** for each cell of a pair of spreadsheets (e.g., the target sheet **14** and the reference sheet **42**) to identify similar spreadsheet regions (e.g., the region **48**, the region **50**) between the pair of spreadsheets.

[0092] At **708**, the method **700** includes providing a first output indicating the pair of spreadsheets are similar or dis-similar. The machine learning model **208** provides a first output (e.g., the vectors **38**) that indicates whether the pair of spreadsheets are similar or dis-similar.

[0093] At **710**, the method **700** includes providing a second output indicating whether the spreadsheets regions are similar or dis-similar between the pair of spreadsheets. The machine learning model **210** provides a second output (e.g., the vectors **40**) that indicates whether the pair of spreadsheets are similar or dis-similar.

[0094] The method **700** may optionally include building an index of similar spreadsheets using the first output indicating the pair of spreadsheets are similar and the second output indicating regions of the pair of spreadsheets are similar. For example, a similar spreadsheet index is created with similar spreadsheets that is used to quickly find similar spreadsheets. Another example includes a similar region index is created with similar spreadsheet regions that can be used to quickly find similar regions. In some implementations, the indexes are approximate nearest neighbor (ANN) indexes. The indexes aid in efficient and accurate identification of similar spreadsheets and/or spreadsheet regions.

[0095] The method **700** may optionally include receiving a target spreadsheet with a target cell; identifying, using the index, a reference spreadsheet that is similar to the target spreadsheet; identifying, using the second machine learning model, a similar region in the reference spreadsheet to the target cell in response to receiving an indication that a formula is being created in the target cell; identifying, using the second machine learning model, a reference formula in the similar region in the reference spreadsheet; automatically modifying, using the second machine learning model, the reference formula with parameters from the target spreadsheet to generate a predicted formula for the target cell; and providing an output in the target cell with the predicted formula.

[0096] The method **700** may optionally include receiving a target spreadsheet; using the index to identify a plurality of similar spreadsheets to the target spreadsheet; and automatically performing a task on the plurality of similar spreadsheets. In some implementations, the task is automatically applying a category to the plurality of similar spreadsheets. For example, thousands of spreadsheets in a company related to travel may be identified in the index as similar and a user may apply a travel category to one of spreadsheets related to travel, the task automatically applies the travel category to the thousands of spreadsheets identified in the index as similar to the spreadsheet that the user labeled as a travel category.

[0097] In some implementations, the task is automatically applying a security label to the plurality of similar spreadsheets. For example, a user may identify a spreadsheet as confidential, and the task automatically applies the same confidential label to the plurality of similar spreadsheets identified in the index as similar to the spreadsheet the user marked as confidential. In some implementations, the task is automatically applying access permissions to the plurality of similar spreadsheets. For example, a user may identify a spreadsheet as being accessible by the human resource department and the task automatically applies the access permission (only the human resource department) to the plurality of similar spreadsheets to the spreadsheet.

[0098] The method **700** may be used to identify similar spreadsheets and/or spreadsheet regions. In addition, the method **700** may be used to aid in data discovery by automatically identifying similar spreadsheets. The method **700** may be used to automatically identify similar spreadsheets and label similar spreadsheets with a category, access permissions, and/or a security level.

[0099] FIG. **8** illustrates an example method **800** for automatically recommending formulas (the predicted formula **16**) for a spreadsheet (a target sheet **14**). The actions of the method **800** are discussed below with reference to FIGS. **2-6**.

[0100] At **802**, the method **800** includes receiving a target spreadsheet with a target cell. The target sheet **14** has a target cell **52**. The target sheet **14** is received in response to the user **212** opening or accessing the target sheet **14**.

[0101] At **804**, the method **800** includes identifying, using a first machine learning model, a reference spreadsheet similar to the target spreadsheet. The machine learning model **208** searches to identify one or more reference sheets in a corpus of spreadsheets that are similar to the target sheet **14**. The machine learning model **208** uses a similar sheet primitive to identify the one or more reference sheets. In some implementations, the search for reference sheet step **214** by the machine learning model **208** occurs automatically in response to the user **212** opening the target sheet **14**. In some implementations, the first machine learning model **208** is trained to identify whether pairs of spreadsheets are similar or dis-similar using training data generated by a weak supervision hypothesis test that compares sheet names of the pair of spreadsheets in determining whether spreadsheets are similar or dis-similar.

[0102] At **806**, the method **800** includes identifying, using a second machine learning model, a similar region in the reference spreadsheet that is similar to the target cell. The machine learning model **210** searches within each reference sheet (e.g., the reference sheet **42**) to identify reference cells (e.g., the reference cell **44**) with local spreadsheet regions (e.g., the region **48**) that look similar to the local region (e.g., the region **50**) around the target cell **52** using a similar region primitive. In some implementations, the second machine learning model **210** is a fully connected network trained to identify whether regions of spreadsheets are similar or dis-similar and trained to identify formulas.

[0103] At **808**, the method **800** includes identifying, using the second machine learning model, a reference formula in the similar region in response to receiving an indication a formula is being created in the target cell. If the reference cell **44** contains a reference formula (e.g., the reference formula **46**), the reference formula is identified by the machine learning model **210** as a likely candidate for a predicted formula **16**. In some implementations, the machine learning model **210** identifies a plurality of reference formulas that are likely candidates for the predicted formula **16** and places the plurality of reference formulas in a ranked order. The machine learning model **210** may take a highest ranked reference formula as the predicted formula **16**. The search for reference formula step **216** occurs in response to the machine learning model **210** receiving an indication that the user **212** is creating a formula in a target cell. One example indication that the user **212** is creating a formula includes the user **212** typing an equal sign in a target cell. Another example indication that the user **212** is creating a formula includes the user **212** typing a partial formula in a target cell. Another example indication that the user **212** is creating a formula includes the user **212** selecting a create formula option.

[0104] At **810**, the method **800** includes modifying, using the second machine learning model, parameters of the reference formula to a context of the target spreadsheet. The machine learning model **210** uses the similar region primitive to adapt the reference formula (e.g., the reference formula **46**) into a local context of the target sheet **14**, by changing the parameter cells of the reference formula. The machine learning model **210** uses the similar-region primitive to learn-to-adapt the parameters from the reference sheet (e.g., the reference sheet **42**) to the target sheet **14**.

[0105] At **812**, the method **800** includes generating, using the second machine learning model, a predicted formula with the modified parameters of the reference formula. The machine learning model **210** generates a predicted formula **16** with parameters from the content of the target sheet **14**.

[0106] At **814**, the method **800** includes providing an output in the target cell with the predicted formula. The output automatically presents the predicted formula **16** in the target cell **52**. In some implementations, the user **212** accepts the predicted formula **16**. In some implementations, the predicted formula **16** is displayed in response to the user **212** typing the equal sign prior to the user **212** typing any additional information for the formula. In some implementations, additional

information is received for the formula being created in the target cell **52** and an updated predicted formula **16** is provided in the target cell **52** in response to the receiving the additional information. For example, the user **212** may continue to type in additional parameters for the formula and the predicted formula **16** is updated in response to the additional parameters provided by the user **212**. [0107] The method **800** is used to automatically predict formulas that users want to author in a target spreadsheet cell.

[0108] FIG. **9** illustrates components that may be included within a computer system **900**. One or more computer systems **900** may be used to implement the various methods, devices, components, and/or systems described herein.

[0109] The computer system **900** includes a processor **901**. The processor **901** may be a general-purpose single or multi-chip microprocessor (e.g., an Advanced RISC (Reduced Instruction Set Computer) Machine (ARM)), a special purpose microprocessor (e.g., a digital signal processor (DSP)), a graphics processing unit (GPU), a microcontroller, a programmable gate array, etc. The processor **901** may be referred to as a central processing unit (CPU). Although just a single processor **901** is shown in the computer system **900** of FIG. **9**, in an alternative configuration, a combination of processors (e.g., an ARM and DSP) could be used.

[0110] The computer system **900** also includes memory **903** in electronic communication with the processor **901**. The memory **903** may be any electronic component capable of storing electronic information. For example, the memory **903** may be embodied as random access memory (RAM), read-only memory (ROM), magnetic disk storage mediums, optical storage mediums, flash memory devices in RAM, on-board memory included with the processor, erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM) memory, registers, and so forth, including combinations thereof.

[0111] Instructions **905** and data **907** may be stored in the memory **903**. The instructions **905** may be executable by the processor **901** to implement some or all of the functionality disclosed herein. Executing the instructions **905** may involve the use of the data **907** that is stored in the memory **903**. Any of the various examples of modules and components described herein may be implemented, partially or wholly, as instructions **905** stored in memory **903** and executed by the processor **901**. Any of the various examples of data described herein may be among the data **907** that is stored in memory **903** and used during execution of the instructions **905** by the processor **901**.

[0112] A computer system **900** may also include one or more communication interfaces **909** for communicating with other electronic devices. The communication interface(s) **909** may be based on wired communication technology, wireless communication technology, or both. Some examples of communication interfaces **909** include a Universal Serial Bus (USB), an Ethernet adapter, a wireless adapter that operates in accordance with an Institute of Electrical and Electronics Engineers (IEEE) 802.11 wireless communication protocol, a Bluetooth® wireless communication adapter, and an infrared (IR) communication port.

[0113] A computer system **900** may also include one or more input devices **911** and one or more output devices **913**. Some examples of input devices **911** include a keyboard, mouse, microphone, remote control device, button, joystick, trackball, touchpad, and lightpen. Some examples of output devices **913** include a speaker and a printer. One specific type of output device that is typically included in a computer system **900** is a display device **915**. Display devices **915** used with embodiments disclosed herein may utilize any suitable image projection technology, such as liquid crystal display (LCD), light-emitting diode (LED), gas plasma, electroluminescence, or the like. A display controller **917** may also be provided, for converting data **907** stored in the memory **903** into text, graphics, and/or moving images (as appropriate) shown on the display device **915**.

[0114] The various components of the computer system **900** may be coupled together by one or more buses, which may include a power bus, a control signal bus, a status signal bus, a data bus, etc. For the sake of clarity, the various buses are illustrated in FIG. **9** as a bus system **919**.

[0115] In some implementations, the various components of the computer system 900 are implemented as one device. For example, the various components of the computer system 900 are implemented in a mobile phone or tablet. Another example includes the various components of the computer system 900 implemented in a personal computer. Another example includes the various components of the computer system 900 implemented in the cloud.

[0116] As illustrated in the foregoing discussion, the present disclosure utilizes a variety of terms to describe features and advantages of the model evaluation system. Additional detail is now provided regarding the meaning of such terms. For example, as used herein, a “machine learning model” refers to a computer algorithm or model (e.g., a classification model, a clustering model, a regression model, a language model, an object detection model, a probabilistic graphical model) that can be tuned (e.g., trained) based on training input to approximate unknown functions. For example, a machine learning model may refer to a neural network (e.g., a convolutional neural network (CNN), deep neural network (DNN), recurrent neural network (RNN)), or other machine learning algorithm or architecture that learns and approximates complex functions and generates outputs based on a plurality of inputs provided to the machine learning model. As used herein, a “machine learning system” may refer to one or multiple machine learning models that cooperatively generate one or more outputs based on corresponding inputs. For example, a machine learning system may refer to any system architecture having multiple discrete machine learning components that consider different kinds of information or inputs.

[0117] The techniques described herein may be implemented in hardware, software, firmware, or any combination thereof, unless specifically described as being implemented in a specific manner. Any features described as modules, components, or the like may also be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a non-transitory processor-readable storage medium comprising instructions that, when executed by at least one processor, perform one or more of the methods described herein. The instructions may be organized into routines, programs, objects, components, data structures, etc., which may perform particular tasks and/or implement particular data types, and which may be combined or distributed as desired in various implementations.

[0118] Computer-readable mediums may be any available media that can be accessed by a general purpose or special purpose computer system. Computer-readable mediums that store computer-executable instructions are non-transitory computer-readable storage media (devices). Computer-readable mediums that carry computer-executable instructions are transmission media. Thus, by way of example, and not limitation, implementations of the disclosure can comprise at least two distinctly different kinds of computer-readable mediums: non-transitory computer-readable storage media (devices) and transmission media.

[0119] As used herein, non-transitory computer-readable storage mediums (devices) may include RAM, ROM, EEPROM, CD-ROM, solid state drives (“SSDs”) (e.g., based on RAM), Flash memory, phase-change memory (“PCM”), other types of memory, other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer.

[0120] The steps and/or actions of the methods described herein may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0121] The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database, a datastore, or another data structure), ascertaining and the

like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing, predicting, inferring, and the like.

[0122] The articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements in the preceding descriptions. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one implementation” or “an implementation” of the present disclosure are not intended to be interpreted as excluding the existence of additional implementations that also incorporate the recited features. For example, any element described in relation to an implementation herein may be combinable with any element of any other implementation described herein. Numbers, percentages, ratios, or other values stated herein are intended to include that value, and also other values that are “about” or “approximately” the stated value, as would be appreciated by one of ordinary skill in the art encompassed by implementations of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable manufacturing or production process, and may include values that are within 5%, within 1%, within 0.1%, or within 0.01% of a stated value.

[0123] A person having ordinary skill in the art should realize in view of the present disclosure that equivalent constructions do not depart from the spirit and scope of the present disclosure, and that various changes, substitutions, and alterations may be made to implementations disclosed herein without departing from the spirit and scope of the present disclosure. Equivalent constructions, including functional “means-plus-function” clauses are intended to cover the structures described herein as performing the recited function, including both structural equivalents that operate in the same manner, and equivalent structures that provide the same function. It is the express intention of the applicant not to invoke means-plus-function or other functional claiming for any claim except for those in which the words ‘means for’ appear together with an associated function. Each addition, deletion, and modification to the implementations that falls within the meaning and scope of the claims is to be embraced by the claims.

[0124] The present disclosure may be embodied in other specific forms without departing from its spirit or characteristics. The described implementations are to be considered as illustrative and not restrictive. The scope of the disclosure is, therefore, indicated by the appended claims rather than by the foregoing description. Changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method, comprising: automatically generating, by a machine learning model, training data with positive examples of similar spreadsheets and negative examples of dis-similar spreadsheets; using the training data, by a first machine learning model, to determine whether a pair of spreadsheets are similar; using the training data, by a second machine learning model, to identify similar spreadsheet regions between the pair of spreadsheets; providing a first output indicating the pair of spreadsheets are similar or dis-similar; and providing a second output indicating whether the spreadsheets regions are similar or dis-similar between the pair of spreadsheets.
2. The method of claim 1, wherein the machine learning model uses a weak supervision hypothesis test to compare sheet names of pairs of spreadsheets in the training data and outputs a positive example in response to the sheet names matching and outputs a negative example in response to the sheet names having different names.
3. The method of claim 2, wherein the machine learning model further uses a probability of finding the sheet names in the training data and outputs the positive example in response to determining the

probability of the sheet names is uncommon in the training data and outputs the negative example in response to determining the probability of the sheet names is common in the training data.

4. The method of claim 1, wherein the machine learning model accesses a large number of spreadsheets to automatically generate the training data.
5. The method of claim 1, wherein the training data is represented as a plurality of feature vectors with embeddings for each cell of the spreadsheets.
6. The method of claim 5, wherein the plurality of feature vectors include a semantic feature vector, a syntactic feature vector, and a style feature vector.
7. The method of claim 5, wherein the plurality of feature vectors are aggregated together into an input vector for each cell of the spreadsheets that represents a region around each cell of the spreadsheet.
8. The method of claim 7, wherein the first machine learning model uses the input vector for each cell to determine whether the pair of spreadsheets are similar and the second machine learning model uses the input vector for each cell to identify the similar spreadsheet regions between the pair of spreadsheets.
9. The method of claim 1, further comprising: building an index of similar spreadsheets using the first output indicating the pair of spreadsheets are similar and the second output indicating regions of the pair of spreadsheets are similar.
10. The method of claim 9, further comprising: receiving a target sheet with a target cell; identifying, using the index, a reference sheet that is similar to the target sheet; identifying, using the second machine learning model, a similar region in the reference sheet to the target cell in response to receiving an indication that a formula is being created in the target cell; identifying, using the second machine learning model, a reference formula in the similar region in the reference sheet; automatically modifying, using the second machine learning model, the reference formula with parameters from the target sheet to generate a predicted formula for the target cell; and providing an output in the target cell with the predicted formula.
11. The method of claim 9, further comprising: receiving a target sheet; using the index to identify a plurality of similar spreadsheets to the target sheet; and automatically performing a task on the plurality of similar spreadsheets.
12. The method of claim 11, wherein the task is automatically applying a category to the plurality of similar spreadsheets.
13. The method of claim 11, wherein the task is automatically applying a security label to the plurality of similar spreadsheets.
14. The method of claim 11, wherein the task is automatically applying access permissions to the plurality of similar spreadsheets.
15. A method, comprising: receiving a target sheet with a target cell; identifying, using a first machine learning model, a reference sheet similar to the target sheet; identifying, using a second machine learning model, a similar region in the reference sheet that is similar to the target cell; identifying, using the second machine learning model, a reference formula in the similar region in response to receiving an indication a formula is being created in the target cell; modifying, using the second machine learning model, parameters of the reference formula to a context of the target sheet; generating, using the second machine learning model, a predicted formula with the modified parameters of the reference formula; and providing an output in the target cell with the predicted formula.
16. The method of claim 15, wherein the output automatically presents the predicted formula in the target cell.
17. The method of claim 15, wherein a user accepts the predicted formula.
18. The method of claim 15, further comprising: receiving additional information for the formula being created in the target cell; and providing an updated predicted formula in the target cell in response to receiving the additional information.

19. The method of claim 15, wherein the first machine learning model is trained to identify whether pairs of spreadsheets are similar or dis-similar using training data generated by a weak supervision hypothesis test that compares sheet names of the pair of spreadsheets in determining whether the pairs of spreadsheets are similar or dis-similar.

20. The method of claim 15, wherein the second machine learning model is a fully connected network trained to identify whether regions of spreadsheets are similar or dis-similar and trained to identify formulas.
